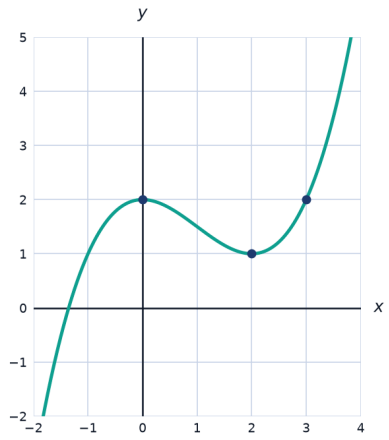
Non-linear Simultaneous Equations (Extended)

Core Concepts: Non-linear Simultaneous Equations (Extended)

1 For the model $f(x) = 2x + 4$ in non-linear simultaneous equations (extended), evaluate $f(2)$.



a. 8

b. 10

c. 4

d. 12

2 Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

3 A data set recorded for non-linear simultaneous equations (extended) consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 2x + 4$ in non-linear simultaneous equations (extended), evaluate $f(3)$.

a. 10

b. 12

c. 6

d. 14

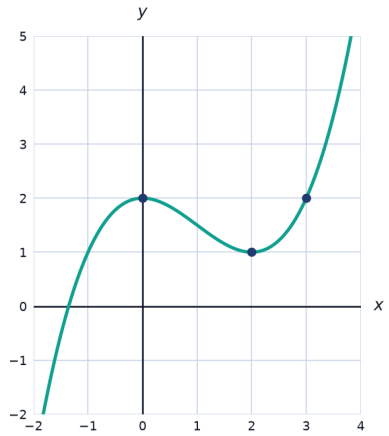
5 Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$



6 A data set recorded for non-linear simultaneous equations (extended) consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 2x + 4$ in non-linear simultaneous equations (extended), evaluate $f(4)$.

a. 12

b. 14

c. 8

d. 16

8 Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

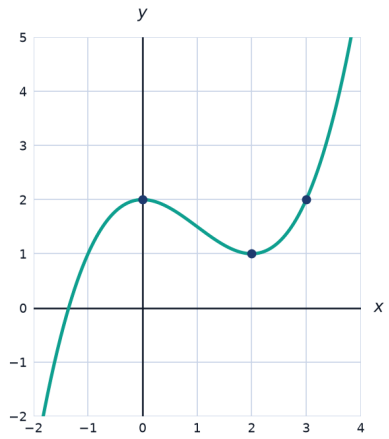
9 A data set recorded for non-linear simultaneous equations (extended) consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

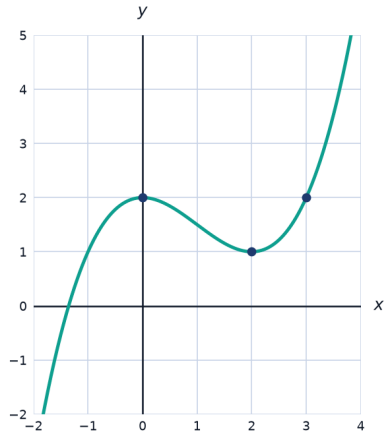
d. 20.5



Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 2x + 4$ in non-linear simultaneous equations (extended), evaluate $f(5)$.



a. 14

b. 16

c. 10

d. 18

11 Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

12 A data set recorded for non-linear simultaneous equations (extended) consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

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